Alimentazioni e masse (3V3, 5V_SYS, GND_A, GND_D, V1P35, V1P4, VBUS1, VDD_SNVS, VIN_SYS) sono etichette globali nei fogli: non sono disegnate qui.

CPU

DDR_A[0..15]
DDR_BA[0..2]
DDR_DQ[0..15]
DDR_CAS_N
DDR_CKE
DDR_CK_N
DDR_CK_P
DDR_CS_N
DDR_LDM
DDR_LDQS_N
DDR_LDQS_P
DDR_ODT
DDR_RAS_N
DDR_RESET_N
DDR_UDM
DDR_UDQS_N
DDR_UDQS_P
DDR_VREF
DDR_WE_N
SOC_DDR_ZQ

DDR3L

DDR_A[0..15]
DDR_BA[0..2]
DDR_DQ[0..15]
DDR_CAS_N
DDR_CKE
DDR_CK_N
DDR_CK_P
DDR_CS_N
DDR_LDM
DDR_LDQS_N
DDR_LDQS_P
DDR_ODT
DDR_RAS_N
DDR_RESET_N
DDR_UDM
DDR_UDQS_N
DDR_UDQS_P
DDR_VREF
DDR_WE_N
SOC_DDR_ZQ

File: ddr3l.kicad_sch

DAC61402_FAULT
ECSPI1_SCLK
ECSPI2_SCLK
MAIN_HIT_N
MAIN_I2C_SCL
MAIN_I2C_SDA
MAIN_SPI_ADDR_0
MAIN_SPI_ADDR_1
MAIN_SPI_ADDR_2
MAIN_SPI_ALERT_N
MAIN_SPI_CS_MASTER
MAIN_SPI_MISO
MAIN_SPI_MOSI
MAIN_SPI_PREFIX_0
MAIN_SPI_PREFIX_1
MAIN_SPI_PREFIX_2
PWR_SPI_DAC_CS
PWR_SPI_INA239A_CS
PWR_SPI_INA239B_CS
PWR_SPI_MISO
PWR_SPI_MOSI
VBIAS_ENABLE
VREF_ENABLE

Bus MIKILAB

DAC61402_FAULT
ECSPI1_SCLK
ECSPI2_SCLK
MAIN_HIT_N
MAIN_I2C_SCL
MAIN_I2C_SDA
MAIN_SPI_ADDR_0
MAIN_SPI_ADDR_1
MAIN_SPI_ADDR_2
MAIN_SPI_ALERT_N
MAIN_SPI_CS_MASTER
MAIN_SPI_MISO
MAIN_SPI_MOSI
MAIN_SPI_PREFIX_0
MAIN_SPI_PREFIX_1
MAIN_SPI_PREFIX_2
PWR_SPI_DAC_CS
PWR_SPI_INA239A_CS
PWR_SPI_INA239B_CS
PWR_SPI_MISO
PWR_SPI_MOSI
VBIAS_ENABLE
VREF_ENABLE

File: bus-mikilab.kicad_sch

I2C1_SCL
I2C1_SDA
PHY_MDC
PHY_MDIO
PHY_RST_N
RMII_CRSDV
RMII_REFCLK
RMII_RXD0
RMII_RXD1
RMII_RXER
RMII_TXD0
RMII_TXD1
RMII_TXEN

Rete

I2C1_SCL
I2C1_SDA
PHY_MDC
PHY_MDIO
PHY_RST_N
RMII_CRSDV
RMII_REFCLK
RMII_RXD0
RMII_RXD1
RMII_RXER
RMII_TXD0
RMII_TXD1
RMII_TXEN

File: rete.kicad_sch

BT_DIS_N
BT_HOST_WAKE
BT_UART_CTS
BT_UART_RX
BT_UART_TX
BT_WAKE_HOST
SD2_CLK_S
SD2_CMD
SD2_D0
SD2_D1
SD2_D2
SD2_D3
WL_DIS_N
WL_WAKE_HOST

Radio

BT_DIS_N
BT_HOST_WAKE
BT_UART_CTS
BT_UART_RX
BT_UART_TX
BT_WAKE_HOST
SD2_CLK_S
SD2_CMD
SD2_D0
SD2_D1
SD2_D2
SD2_D3
WL_DIS_N
WL_WAKE_HOST

File: radio.kicad_sch

GNSS_EXTINT
GNSS_PPS
GNSS_UART_RX
GNSS_UART_TX

GNSS

GNSS_EXTINT
GNSS_PPS
GNSS_UART_RX
GNSS_UART_TX

File: gnss.kicad_sch

USB1_DM
USB1_DP
USB2_DM
USB2_DP
USB2_OC_N
USB2_VBUS_EN

USB

USB1_DM
USB1_DP
USB2_DM
USB2_DP
USB2_OC_N
USB2_VBUS_EN

File: usb.kicad_sch

QSPI_CS1_N
QSPI_CS_N
QSPI_D0
QSPI_D1
QSPI_D2
QSPI_D3
QSPI_SCLK_S
SD1_CD_N
SD1_CLK_S
SD1_CMD
SD1_D0
SD1_D1
SD1_D2
SD1_D3

Storage

QSPI_CS1_N
QSPI_CS_N
QSPI_D0
QSPI_D1
QSPI_D2
QSPI_D3
QSPI_SCLK_S
SD1_CD_N
SD1_CLK_S
SD1_CMD
SD1_D0
SD1_D1
SD1_D2
SD1_D3

File: storage.kicad_sch

POR_B
PWR_SRC_ST

Alimentazione

POR_B
PWR_SRC_ST

File: alimentazione.kicad_sch

File: cpu.kicad_sch

Sheet: /
File: SchedaIMX6ULL-SD.kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/10

U1 NXP MCIMX8YZCVM09AB (LMX 6GLL full feature, 792 MHz, −40..105 C. MAPBGA-289 14x14x mm passo 0.8). Stevo: IMX8ULLCEC Rev 1.2 tabelle 90/91, confrontate con la netlist di una scheda KiCad aperta: 289/289.

Boot: BOOT_MODE[1:0]=10 (boot interno) e BOOT_CFG tutti bassi con i pull-down interni da 100k (tabella 78) = QSPI, come SW001 tutto OFF sulla EVK NXP. Pull-up 2K2 su LCD_DATA04 e LCD_DATA06 (micro50). Se QSPI e' vuota il ROM va in download seriale su USB_OTG1.

Assegnazione: ECSPI2=BOOT_POWER, ECSPI1=MAIN_I2C2=bus MIKROLAB, I2C1=EEPROM MAC, UART1=console, UART3=GNSS, UART4=BT, GPT1 CAPTURE1=PPS (UART2_TX), CAPTURE2=EXTINT (UART2_RX). Elenco completo in docs/pinmap-imx8ull.md.

CSIDM_A[0..15]
CSIDM_D[0..15]
CSIDM_BA[0..2]

U1A
MCIMX8YZCVM08AB
DDR1

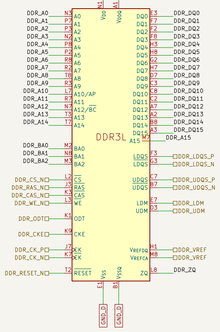
DDR_A0
DDR_A1
DDR_A2
DDR_A3
DDR_A4
DDR_A5
DDR_A6
DDR_A7
DDR_A8
DDR_A9
DDR_A10
DDR_A11
DDR_A12
DDR_A13
DDR_A14
DDR_A15
DDR_CAS_N
DDR_CS_N0
DDR_DQ0
DDR_DQ1
DDR_DQ2
DDR_DQ3
DDR_DQ4
DDR_DQ5
DDR_DQ6
DDR_DQ7
DDR_DQ8
DDR_DQ9
DDR_DQ10
DDR_DQ11
DDR_DQ12
DDR_DQ13
DDR_DQ14
DDR_DQ15
CSIDM_CSN
CSIDM_ODM
CSIDM_DST
CSIDM_RAS_N
CSIDM_RESET_N
CSIDM_WE_N
CSIDM_BA0
CSIDM_BA1
CSIDM_CKE
CSIDM_CK_N
CSIDM_CK_P
CSIDM_LDQS_N
CSIDM_LDQS_P
CSIDM_UDQS_N
CSIDM_UDQS_P
CSIDM_CAS_N
CSIDM_WE_N
CSIDM_CKE1
CSIDC_DVR_EN

U1B
MCIMX8YZCVM08AB
Alimentazioni

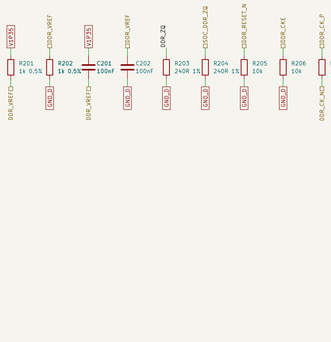
BOOT_MODE0
DTACK
VDD_ARM_CAP
VDD_ARM_CAP
VDD_ARM_CAP

U2 Micron MT41K512M16HA-107:A. 8 Gb x16 = 1 GB. FBGA-96 9x14 passo 0.8. a 1.35 V. Un solo chip punto-punto: niente VTT. Simbolo da MT41K256M16HA di KiCad con A15 su M7 (ball map JEDEC x16).

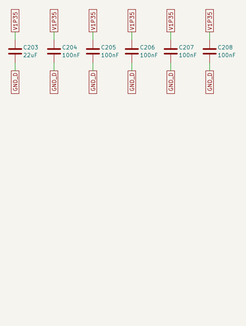
VREF a partitore 1k 0.5% con 100 nF su ogni resistenza (datasheet 6ULL, tabella 3), unica per DRAM_VREF, VREFCA, VREFDQ. ZQ 240R 1% su SoC e DRAM. RESET# e CKE con 10k a GND. CK con 100R differenziale vicino alla DRAM (da confermare in simulazione).



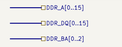
VREF, ZQ, pull-down



VDD / VDDQ 1.35 V







Sheet: /DDR3L/		
File: ddr3l.kicad_sch		
Title:		
Size: A1	Date:	**Rev:**
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VIN_SYS 12 V dal BUS_POWER -> TPS56637 5 V (U22) -> TPS2121 U17 (IN1 bus con priorità, IN2 VBUS USB-C 1) -> 5V_SYS -> buck 3.3 V (U23), 1.35 V DDR (U26), 1.4 V VDD_SOC_IN (U27).

Sequenza 3.3 -> 1.35 -> 1.4 V a RC sugli EN del TLV62569 (soglia EN max 1.2 V): 100k/470nF circa 21 ms, 100k/1uF circa 45 ms. POR_B dal TPS3839K33 (2.93 V, push-pull) con 1k in serie, così SW1 può cortocircuitare POR_B.

VDD_SNVS: OR a diodi BAT54C tra 3V3 e CR1220 (BT1), alimenta anche V_BCKP del GNSS. Nessuna ricarica della pila. Passivi 0603 minimo: 47 uF e 10 uF/25 V in 1206, 22 uF e 10 uF in 0805.

TPS56637: EN 169k/36.1k, FB 73.2k/10k = 4.99 V, FF 20k+100pF, MODE aperto = FCCM, 2x47 uF ai posto di 4x22 uF

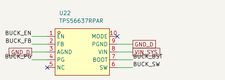
TPS2121 (come RK3568): PR1 4.49 V, OV 5.98 V, ILM 22.1k = 4–5 A, SS 100 nF

U23 3.3 V / 3 A: FB 33.2k/10k = 3.32 V, primo della sequenza

U26 1.35 V DDR: FB 150k/120k (0.6 V x 2.25), EN 100k/470nF da 3V3

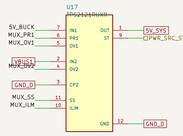
U27 1.4 V VDD_SOC_IN: FB 133k/100k = 1.398 V (min 1.375 V, 125 mV sopra VDD_ARM_CAP), EN 100k/1uF da 3V3

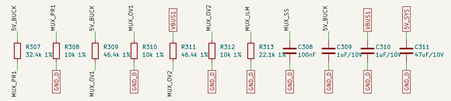
POR_B

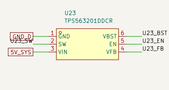












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KiCad E.D.A. 10.0.7 | Id: 4/10

J11 BUS_POWER 30 pin (Hirose FH12-30S) su ECSPI2. J12 MAIN 28 pin REV5.1 (XUNPU FPC-05F-28PH20, pin 28 = HIT#) su ECSPI1. Mappa del connettori identica a SchedaA133/RK3568. Rail BUS_* passanti, non collegati ai rail della scheda.

Serie 33R solo sul clock SPI. Pull-up 4.7k su ALERT#/HIT#, 2.2k su I2C. VREF_ENABLE 10k e VBIAS_ENABLE 100k a GND. CS e FAULT con i pull-up interni del 6ULL (pad control nel device tree). SPI_ID a GND diretti. ESD TPD4E05U06DQA sulle linee logiche.

J11
BUS_POWER_MIKILAB

J12
BUS_MIKILAB_MAIN

Serie SPI 33R, solo clock

Livelli di riposo

U29
TPD4E05U06DQAR

U30
TPD4E05U06DQAR

U31
TPD4E05U06DQAR

U32
TPD4E05U06DQAR

U33
TPD4E05U06DQAR

U34
TPD4E05U06DQAR

Disaccoppiamento connettori

#FLG450
PWR_FLAG

#FLG451
PWR_FLAG

#FLG452
PWR_FLAG

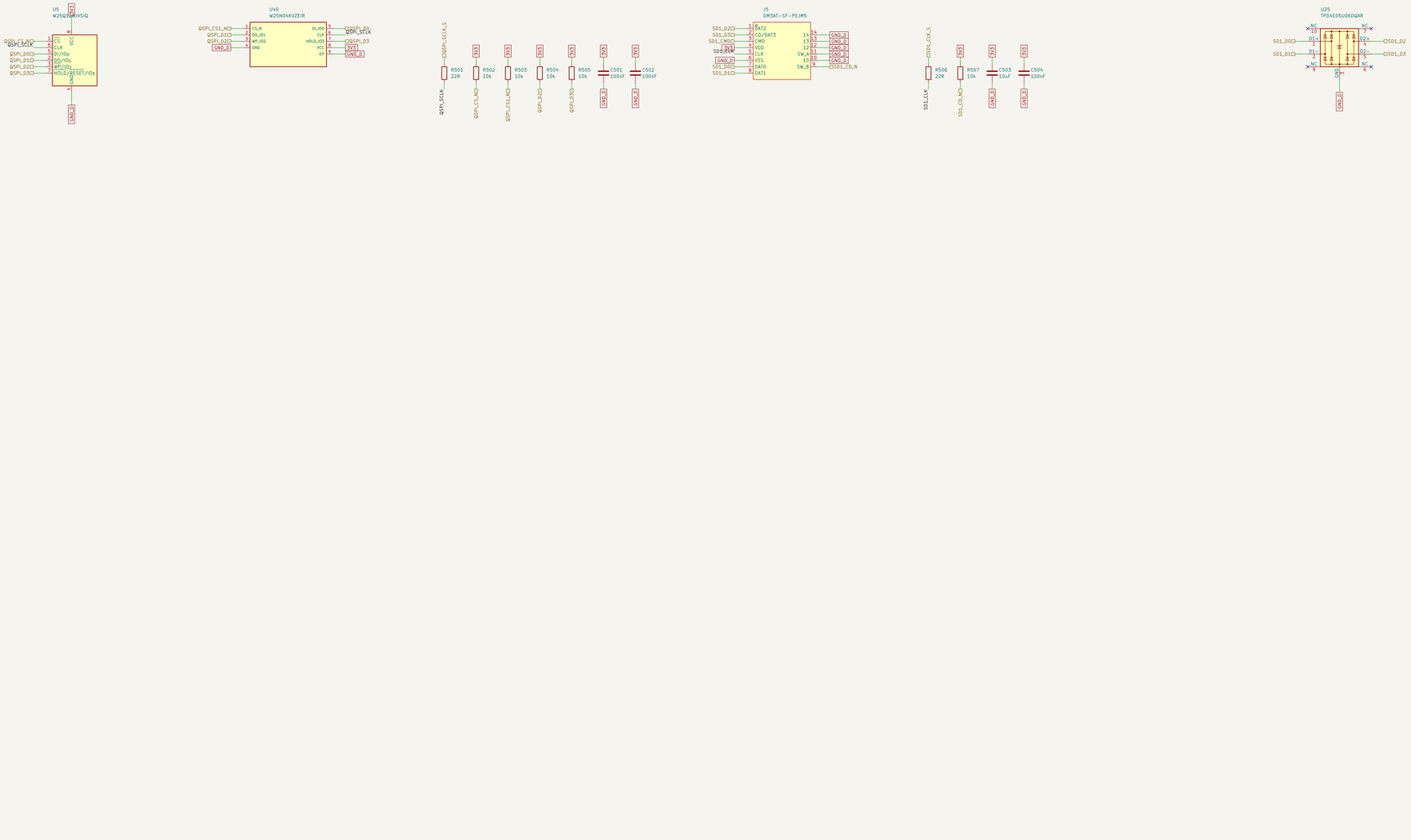
Sheet:	/Bus MIKILAB/
File:	bus-mikilab.kicad_sch
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Size:	A1
Date:	
Rev:	
KiCad E.D.A.	10.0.7
Id:	5/10

U5 W25Q128JV 16 MB su QSPI1 porta A, SS0: boot. U40 W25N04KV 512 MB SPI NAND SLC su SS1 (NAND_DATA07): rootfs UBIFS. Bus QSPI in comune. Il ROM del 6ULL non avvia da SPI NAND: per questo restano due chip.

J5 microSD DM3AT su uSDHC1 (sul BOTTOM). 3.3 V. CD su UART1_RTS_B (USDHC1_CD_B). U25 protegge DAT0–3: CMD e CLK non hanno ESD (un solo TPD4E in BOM, 4 linee).

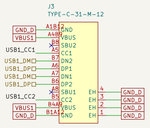
QSPI: 22R su SCLK, pull-up 10k su CS0/CS1/IO2/IO3

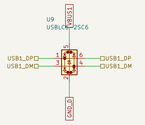
microSD: 22R su CLK, pull-up 10k su CD; CMD e DAT con i pull-up interni del 6ULL



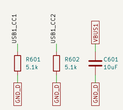
J3 USB-C 1: USB_OTG1, device/download seriale e alimentazione (VBUS1 -> IN2 del TPS2121). Sink: 5,1k su CC1/CC2. J8 USB-C 2 host: USB_OTG2 diretto, VBUS dal TPS2553 (~1 A). Rp 56k. Le due USB-C sono affiancate sullo stesso bordo.

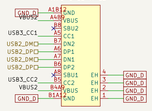
USB_OTG2_VBUS del 6ULL da 5V_SYS. Niente hub: J8 e' l'unica porta host.

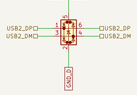




CC sink J3

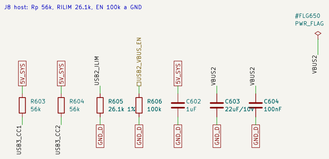








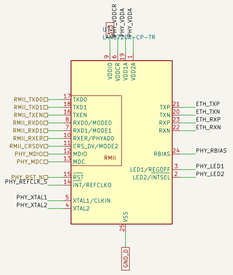
J8 host: Rp 56k. RILIM 26.1k. EN 100k a GND

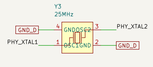


Sheet: /USB/		
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Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 7/10

U7 LAN8720A su ENET1 RMII. Y3 25 MHz montato: il PHY genera i 50 MHz (nINTSEL basso) su REFCLK0 verso ENET1_TX_CLK (REF_CLK in ingresso, SION). nINT quindi non disponibile. Strap MODE=111 (10k su RXD0/RXD1/CRS_DV). PHYADD=0.

J1 HR911105A: magnetici e LED integrati. TX/RX con 49.9R a VDDA. CT a VDDA con 22 nF. Polarita' del LED dal disegno Hanrun: da verificare (solo indicazione, non blocca il link). U6 24AA02E48 su I2C1 con il MAC EUI-48.

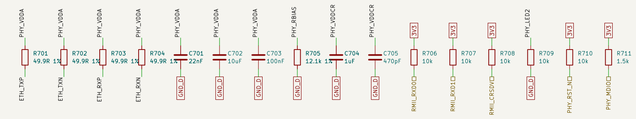


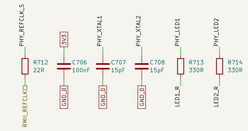




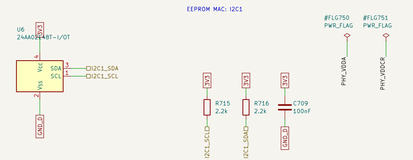


PHY: terminazioni, RBIAS 12.1k, VDDCR 1uF+470pF, straps, REFCLK 22R





EEPROM MAC: I2C1

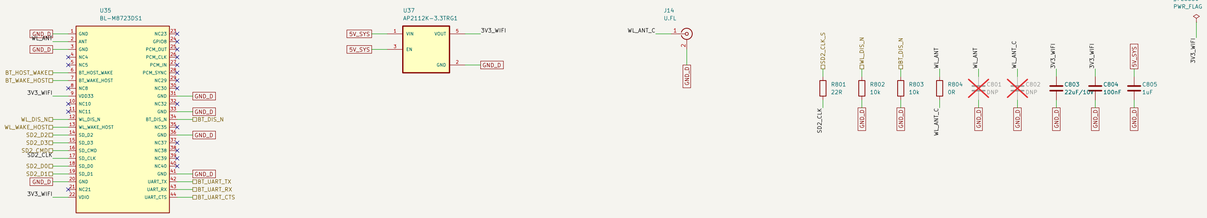


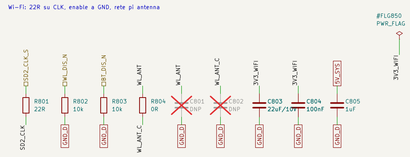
Sheet: /Rete/
File: rete.kicad_sch
Title:
Size: A1 | Date: | Rev:
KiCad E.D.A. 10.0.7 | Id: 8/10

U35 LB-LINK BL-M8723DS1 (RTL8723DS, 12x12 mm, datasheet V1.0.1.3): Wi-Fi su uSDHC2 (pad NAND), BT su UART8 (pad ENET2). VDD33 e VDIO da U37 (3.3 V dedicati ai picchi TX). WL_DIS# e BT_DIS# senza pull-up nel modulo: 10k a GND, li alza Linux.

SDIO con i pull-up interni del 6ULL. Antenna su U.FL J14 (scelta del placement: la BOM lasciava u.FL o PCB) con rete pi OR + 2 DNP da tarare. UART_CTS del modulo da UART8_RTS. Il modulo non ha RTS (pin 41 = GND): H5 a 3 fili. PCM non usata: PCM_OUT libero (strap da non alzare).

Wi-Fi: 22R su CLK, enable a GND, rete pi antenna

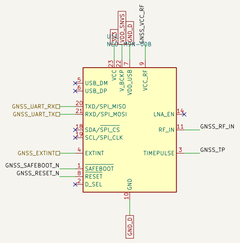




Sheet: /Radio/		
File: radio.kicad_sch		
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Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 9/10

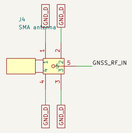
U11 u-blox NEO-M9N-00B su UART3. TIMEPULSE -> 33R -> GNSS_PPS su GPT1_CAPTURE1 (UART2_TX) e buffer U20 -> 33R -> header J7. EXTINT IN da J7 -> ESD + 100R -> buffer U19 -> GNSS_EXTINT su NEO e GPT1_CAPTURE2 (UART2_RX): evento marcato da entrambi.

J4 SMA antenna, bias 3.3 V attiva: VCC_RF -> 10R -> 27 nH (0603) -> RF_IN, 10 nF. Linea 50 ohm corta. V_BCKP da VDD_SNVS (CR1220 via BAT54C). J7: PPS OUT. EXTINT IN, GND (come scheda Tibbo).



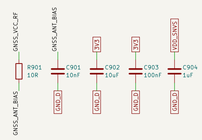


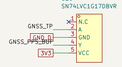






Bias antenna e alimentazione







Timing: serie e buffer



